

Global impacts of a warming world

Climate science consistently shows that limiting the rise in global average surface temperature will help avoid some of the most dangerous impacts of climate change – including limiting flooding from sea level rise, impacts on human health, loss of biodiversity, and many other factors.

Here we use analysis of the Met Office's Earth system model, HadGEM2-ES, to show how some impacts differ at certain levels of warming. The results are taken for a single high emissions scenario as global temperatures pass 1.5 °C, 2 °C and 4 °C of warming above pre-industrial levels. Information on other factors that influence real world responses to climate change, such as the extent to which adaptation measures are undertaken by affected countries, is not included.

Extreme heat: Increases in daily maximum temperature can affect people's ability to work, increase mortality rates within a given population and impact on infrastructure – such as slowing train operating speeds and reducing the ability of energy infrastructure to meet demand.

At a global annual average warming of 4 °C, the increase in the maximum temperatures experienced over some land areas will be, at times, much higher than this. For example, in Central Europe the annual maximum temperature could be around 8 °C above the annual maximum typical of pre-industrial times. Limiting global average surface warming to 2 °C could reduce Central Europe maximum annual temperatures by more than 4 °C, thereby avoiding a large proportion of the associated negative impacts on society. Limiting global warming further to 1.5 °C could avoid more than 1 °C further of the annual maximum in this region. The modelled evidence is clear: reducing global mean warming helps to avoid the likelihood of annual temperature extremes and, thus, a significant proportion of negative regional impacts.

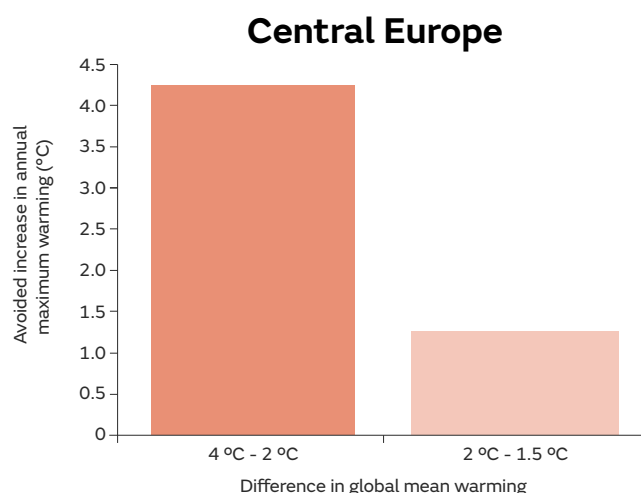


Figure 1. HadGEM2-ES shows the Central Europe annual maximum temperature increases avoided at 2 °C instead of 4 °C, then again at 1.5 °C over to 2 °C under an RCP8.5 scenario.

Droughts and floods: River flows are an important factor to consider because reduced flows can increase the risk of water stress and drought, while increased flows could increase the risk of flooding. While river flows at 4 °C above pre-industrial levels are projected to increase by 12-15% globally, there are large regional variations which suggest impacts could vary greatly, with some regions showing an increase and some a decrease. The map below (left panel) shows limiting warming to 2 °C compared to 4 °C avoids substantial changes in river flows and runoff. When limiting warming to 1.5 °C (right panel) the broad features remain consistent and the impacts are again reduced, but by a smaller amount.

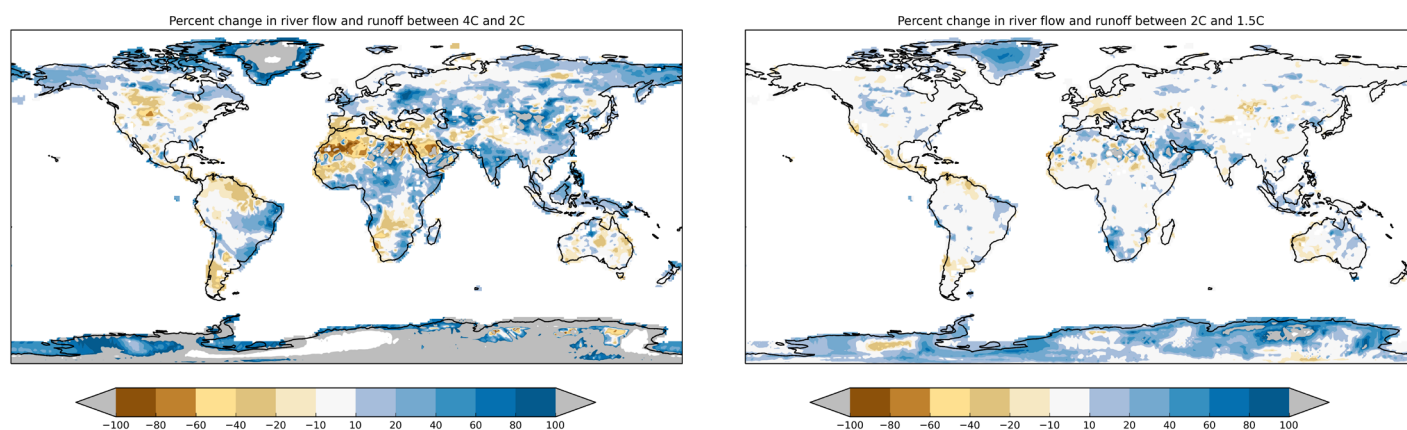


Figure 2. Shows the percentage change in river flows from 2 °C to 4 °C (left panel) and then from 1.5 °C to 2 °C (right panel). It clearly shows mitigation can avoid substantial changes as well as their associated impacts.

Land ecosystems and agriculture: A warming world combined with changes in the amount of carbon dioxide in the atmosphere (with increases potentially enabling faster growth of vegetation) presents a more complex picture for future impacts on crop yields and ecosystems such as forests, as well as their associated biodiversity. Increasing temperatures coupled with this CO₂ 'fertilisation' are expected to continue to extend the growing season across temperate regions. Indeed this has already led to 'global greening' in some areas and, while this is expected to continue in future, other factors are likely to offset this benefit.

- Shifts in rainfall patterns could have adverse effects on plant growth.
- Fire risk is likely to increase with temperatures, which may lead to forest losses which models do not yet account for.
- Land use and land cover are also likely to change depending on how the world decides to reduce emissions (e.g. growth of bioenergy crops coupled with carbon capture and storage, to draw-down atmospheric carbon).
- Crops are likely to be affected by issues such as higher temperatures which increase evaporation and dry out soil in some regions – as shown in the figures below.
- Regional reductions in rainfall and the potential for temperatures to pass critical thresholds for crops are other threats that can reduce yields dramatically.
- Changes in pests and diseases can also be expected, but have not yet been factored into current models.

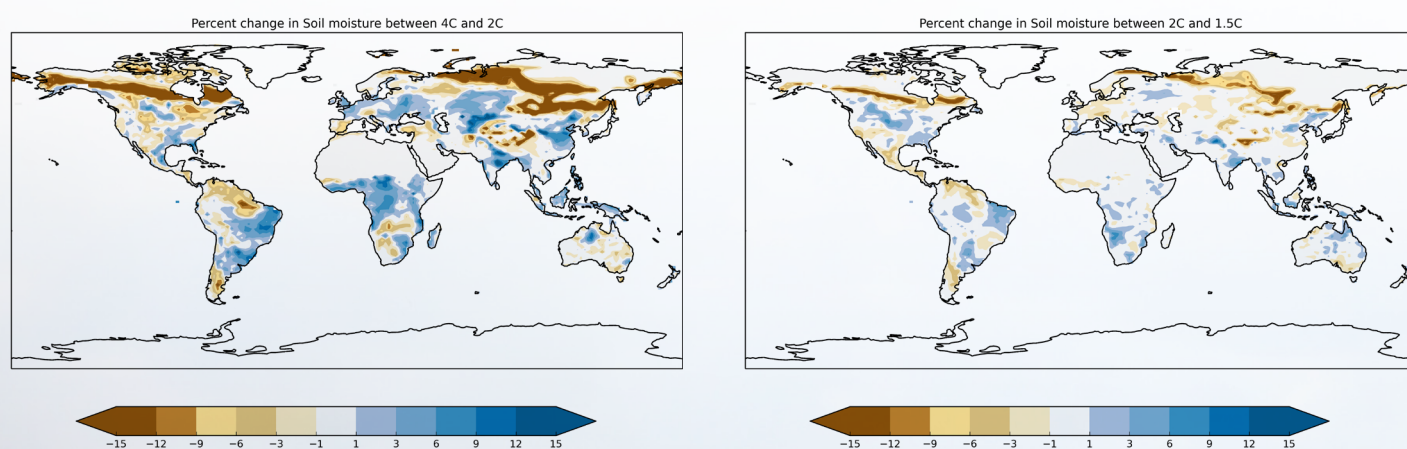


Figure 3. This shows the percentage change in soil moisture for a change in temperature from 2 °C to 4 °C (left panel) and 1.5 °C to 2 °C (right panel).

